To’la differensial tenglamalar. Integrolvchi ko’paytuvchi va uning mavjudligi haqida teoremalar.

Agar $M(x,y)dx+N(x,y)dy=0$ ko‘rinishdagi tenglamaning chap qismi biror $u(x,y)$ funksiyaning to‘liq differensiali, ya’ni $du=M(x,y)dx+N(x,y)dy=0$ *(1)* bo‘lsa, u holda bunday tenglama to‘liq differensialli tenglama deyiladi. *(1)* tenglama to‘liq differensialli tenglama bo‘lishi uchun $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$ shart bajarilishi kerak.

To‘la differensialli tenglama ta’rifidan $du=0$, bundan $u(x,y)=C$ *e*kanligi kelib chiqadi (C- ixtiyoriy o‘zgarmas).

u(x,y) ni toppish uchun *u* ni o'zgarmas deb hisoblaymiz, u holda *dy=0* ekanidan *du=M(x,y)dx* bo'ladi. Bu tenglikni *x* bo'yicha integrallasak,

$$u=\int M(x,y)dx+\varphi(y)\ (2).$$

(2) Tenglikni u bo'yicha differensiallaymiz va natijani N(x,y)ga tenglaymiz, chunki $\frac{\partial u}{\partial y}=N(x,y)$

$$\int \frac{\partial M}{\partial y}dx+\ \varphi'(y)=N(x,y)\quad yoki$$

$$\varphi'(y)=N(x,y)-\int \frac{\partial M}{\partial y}dx\ (3)$$

(3) ifodani u bo'yicha integrallab, s(u) ni topamiz.

$$\varphi(y)=\int(N(x,y)dx-\int \frac{\partial M}{\partial y}dx)dy+C$$

Demak,

$u(x,y)=\int M(x,y)dx+\int(N(x,y)dx-\int\frac{\partial M}{\partial y}dx)dy$

+C.

Bu ifodani o’zgarmasga tenglab, tenglamaning umumiy integralini hosil qilamiz.

1-misol. $(3x^2+6x\,y^2)dx+(6\,x^2y+4\,y^3)dy=0$ tenglamaning umumiy yechimini toping.

Yechish. Bu yerda

M= $3x^2+6x\,y^2$

N= $6\,x^2y+4\,y^3$

$\frac{\partial M}{\partial y}=12xy$, $\frac{\partial M}{\partial y}=12xy$,

Yani $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$.

$\frac{\partial u}{\partial x} = M(x, y)$ bo'lganligi sababli

$$\frac{\partial u}{\partial x} = 3x^2 + 6x\, y^2.$$

Bu tenglik bo'yicha integrallaymiz:

$$u = x^3 + 3x^2\, y^2 + \varphi(y).$$

Bundan $\frac{\partial u}{\partial y} = 6x^2 y + \varphi'(y).$

$\frac{\partial u}{\partial y} = N(x, y)$ ekanligini hisobga olsak,

$$\varphi'(x) = 6x^2 y + 4y^3 - 6x^2 y,$$

Yoki $\varphi'(x) = 4y^3.$

Bundan, $\varphi(x) = y^4 + C$

Javob:

Demak,

$u = x^3 + 3x^2\, y^2 + y^4 + C$

Yoki

$x^3 + 3x^2\, y^2 + y^4 = C$

Integrallovchi ko‘paytuvchi.

Agar $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ bo‘lsa, u holda ba’zi bir shartlar bajarilganda, shunday $\mu(x, y)$ funksiyani topish mumkinki, μMdx+ μNdy=du bo‘ladi. Bu $\mu(x, y)$ funksiya integrallovchi ko‘paytuvchi deyiladi.

Quyidagi hollarda integrallovchi ko‘paytuvchini topish oson:

1) $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \varphi(x)$ bo’lganda,

$ln\ \mu = \int \varphi(x)dx$ bo’ladi.

2) $\frac{\frac{\partial N}{\partial X} - \frac{\partial M}{\partial Y}}{M} = \varphi_1(y)$ bo’lganda

$ln\ \mu = \int \varphi_1(y)dx$ bo’ladi.

2-misol: $(y+xy^2)dx-xdy=0$ tenglamani eching.

Yechish: bu yerda $M= y+xy^2$, $N=-x$

$$\frac{\partial M}{\partial y}=1+2xy, \quad \frac{\partial N}{\partial x}=-1, \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

Demak, tenglamaning chap tomoni biror funksiyaning to‘la differensiali emas. Bu tenglamaning faqat y ga bog‘liq bo‘lgan integrallovchi ko‘paytuvchisi bormi degan masalani qaraymiz.

$$\frac{\frac{\partial N}{\partial X}-\frac{\partial M}{\partial Y}}{M}=\frac{-1-1-2xy}{y+xy^2}=-\frac{2}{y}$$

Bundan,

$ln\ \mu=-2\ lny$, yani $\mu=\frac{1}{y^2}$

Berilgan tenglamaning μ ga ko‘paytirganda keyin

$$(\frac{1}{y}+x)dx-\frac{x}{y^2}dy=0$$

tenglama hosil bo‘ladi. Bu to‘la differensialli tenglamadir, chunki

$$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}=-\frac{1}{y^2}$$

Tenglamani yechib

$$\frac{x}{y}+\frac{x^2}{2}+C=0$$

Yoki $y=-\frac{2x}{x^2+2C}$

Umumiy integralni topamiz.

M va N funksiyalar ikkalasi bir vaqtda nolga teng bo'lmagan funksiyalar, ya'ni $M^2+N^2 \neq 0$ bo'lsin. Agar M va N uzluksiz funksiya bo'lib, uzluksiz xususiy hosilalarga ega bo'lsa, u holda integrallovchi ko'paytuvchi mavjud bo'ladi.

Agar m_0(x,y) funksiya M(x,y)dx + N(x,y)dy = 0 (1) tenglamaning integrallovchi ko'paytuvchisi bo'lib, u_0(x,y) esa (1) tenglamaning shu integrallovchi ko'paytuvchiga mos integrali, ya'ni

$$m_0(\mathrm{Mdx} + \mathrm{Ndy}) = \mathrm{d}u_0$$

Tenglik o'rinli bo'lsa, u holda ixtiyoriy φ differensiallanuvchi funksiya uchun m = m_0(x,y)$\varphi(u_0)$ funksiya ham berilgan tenglamaning integrallovchi ko'paytuvchisi bo'ladi.

Integrallovchi ko'paytuvchining bu xossasi ko'p hollarda berilgan tenglamani ikki qismga ajratish usuli bilan integrallovchi ko'paytuvchini topish imkonini beradi.

M(x y)dx + N(x y)dy =0 tenglama uchun m = m(x y) integrallovchi ko'paytuvchini topish

J A D V A L I

M va N funksiyalarga shartlar	Integrallovchi ko'paytuvchi
$M = y\varphi(xy), \quad N = x\psi(xy)$	$m = 1/(xM - yN),$
$M_x = N_y, \quad M_y = -N_x$	$m = 1/\left(M^2 + N^2\right)$
$\dfrac{M_y - N_x}{N} = \varphi(x)$	$m = \exp\left[\int \varphi(x)\,dx\right]$
$\dfrac{M_y - N_x}{M} = \varphi(y)$	$m = \exp\left[-\int \varphi(y)\,dy\right]$
$\dfrac{M_y - N_x}{N - M} = \varphi(x+y)$	$m = \exp\left[\int \varphi(z)\,dz\right],$ $z = x + y$
$\dfrac{M_y - N_x}{yN - xM} = \varphi(xy)$	$m = \exp\left[\int \varphi(z)\,dz\right], z = xy$
$\dfrac{x^2(M_y - N_x)}{yN + xM} = \varphi\left(\dfrac{y}{x}\right)$	$m = \exp\left[-\int \varphi(z)\,dz\right], \; z = \dfrac{y}{x}$
$\dfrac{M_y - N_x}{xN - yM} = \varphi\left(x^2 + y^2\right)$	$m = \exp\left[\dfrac{1}{2}\int \varphi(z)\,dz\right],$ $z = x^2 + y^2$
$M_y - N_x = \varphi(x)\,N - \psi(y)\,M$	$m = \exp\left[\int \varphi(x)\,dx + \int \psi(y)\,dy\right]$
$\dfrac{M_y - N_x}{N\omega_x - M\omega_y} = \varphi(\omega)$	$m = \exp\left[\int \varphi(\omega)\,d\omega\right]$

Mavzuni yana

…

shu yerdagi misollar orqali mustahhkamlab olamiz.

E’tiboringiz uchun rahmat!